

**LIGHTHIEF**

Energy Storage Solutions | Cyprus

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REQUEST FOR INFORMATION

Comprehensive EMS / SCADA / PPC Technical & Commercial Questionnaire

Cyprus BESS Portfolio — 51 Parks | 249 MW | 882 MWh

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1. INTRODUCTION & PORTFOLIO OVERVIEW

Lighthief Cyprus Ltd is the EPC contractor, exclusive OEM distributor, and long-term O&M service provider for a portfolio of Battery Energy Storage System (BESS) projects across Cyprus. We are seeking proposals from qualified EMS (Energy Management System) providers for the supply, integration, and ongoing support of EMS, PPC (Power Plant Controller), RTU, and SCADA systems across the entire portfolio.

Portfolio Summary:

- **51 Parks** across 5 districts (Nicosia, Famagusta, Limassol, Larnaca, Paphos)
- **249 MW** total power capacity
- **881.78 MWh** total energy capacity
- **251 containers** (66 MV Skids + 185 BESS containers)
- **3 client groups + 6 standalone parks**
- **Target deployment:** Q2 2026 – Q4 2027

1.1 BESS Equipment (OEM Supplied)

Component	Manufacturer	Model / Specification	Communication
Battery Containers	Linyang Energy	Power Atlantic 5.015 MWh, 20HC, LFP (EVE 314Ah), Liquid Cooling	Modbus TCP, CAN, RS485
PCS (Inverter)	Kehua Tech	BCS-C-HUD series (1000K / 1250K / 2500K), EN 50549-2 certified	Modbus TCP, IEC 61850, IEC 104
BMS	Linyang (3-level)	BMU (L1) / BCMU (L2) / BAMS (L3) — 96 BMUs per container	CAN, RS485, Modbus TCP
MV Transformer	Kehua	SL-10000 series, Oil-immersed, 0.69/35kV, Dy11-y11	Modbus, condition monitoring
Switchgear / RMU	Schneider (via Linyang)	SF6 or Air-insulated, 11kV / 22kV	IEC 61850, Modbus

1.2 Regulatory Context

Authority	Role	Key Requirement
EAC (Electricity Authority of Cyprus)	Distribution System Operator (DSO)	IEC 60870-5-104 mandatory, grid connection ≤8MW per park
TSOC	Transmission System Operator (TSO)	Grid connections >8MW, ancillary services market
CERA	Energy Regulatory Authority	Market licensing, tariff structure, NIS2 compliance

1.3 Portfolio Structure

Group	Parks	MW	MWh	Containers	Aggregator
G1: ABIO Power	25	125.0	430.3	125	ABIO
G2: Esperia Energy	11	79.5	310.5	79	Synenergia
G3: Timotheos	9	25.5	81.0	27	TBC
Standalone	6	19.0	60.0	20	Various
TOTAL	51	249.0	881.8	251	

1.4 Site Size Categories

Category	Capacity Range	PCS Units	MV Skids	Parks (approx.)
Small	1–3 MW / 3–10 MWh	1–2	1	~20
Medium	4–8 MW / 10–25 MWh	3–6	1–2	~18
Large	10–15 MW / 35–40 MWh	8–10	2	~8
XL	20–30 MW / 60–100 MWh	15–23	2–3	~5

2. SYSTEM ARCHITECTURE & HARDWARE REQUIREMENTS

2.1 Per-Site Architecture

Required Architecture per Park:

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Cyprus DSO (EAC)
|
| IEC 60870-5-104 (TCP/IP, Port 2404)
|
[EMS / PPC / RTU Controller] - Central site controller
|
|--- Modbus TCP ---> Kehua PCS (BCS-C-HUD) - P/Q setpoints
|--- Modbus TCP ---> Linyang BMS (BAMS Level 3) - battery monitoring
|--- IEC 61850 / Modbus ---> Protection Relays, MV Switchgear
|--- EMS Software - Dispatch, optimization, grid services
|
SCADA Local (per park) ---> SCADA Global (per group / fleet)
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2.2 Hardware Scope — Please Specify What Is Included

Component	Priority	Your Offer (included / excluded / optional)
PPC / RTU / EMS Controller (Combined or separate hardware)	REQUIRED	Included — Software-defined PPC running on NVIDIA Jetson Orin NX (or AGX for XL parks). Combines EMS + PPC + RTU functions in a single edge compute node running scadaServer + commServer. No separate dedicated PPC controller hardware required.
I/O Modules (Digital/analog I/O, power measurement inputs)	REQUIRED	Included — Configurable digital/analog I/O mapping via commServer. Supports Modbus TCP/RTU register mapping for meters, sensors, and relay status. Customer-supplied I/O hardware; driver and configuration included.
Energy Meters (Grid-side, BESS-side power metering)	REQUIRED	Excluded (customer-supplied). Modbus TCP/RTU integration driver included for all major meter brands. Grid-side and BESS-side metering data ingested and displayed in SCADA.
Network Switch (managed)	REQUIRED	Excluded (customer-supplied). Network architecture specification and VLAN configuration guide provided. Compatible with Cisco IE, Moxa EDS, or Hirschmann series managed switches.

Industrial Router / VPN Firewall	REQUIRED	Excluded (customer-supplied). WireGuard/IPSec configuration guide provided. Recommended: Tosibox Lock, Moxa EDR-810, or Fortinet FortiGate Rugged. Dual-WAN failover configuration supported and documented.
4G/LTE Modem / SIM Router	REQUIRED	Excluded (customer-supplied). Dual-WAN failover (Ethernet primary + 4G backup) supported with $\leq 5s$ automatic switchover. No session interruption to IEC-104 DSO link on path failover.
Control Cabinet / Enclosure	PREFERRED	Optional — Pre-integrated DIN-rail panel with edgeServer, switch, and terminal blocks available on request. The premium heavy-duty audioPMD edgeServer in IP67 enclosure option is available for outdoor/container installation. Premium version features embedded Out-of-Band management w. redundant 4G/LTE module.
UPS for Controls	PREFERRED	Excluded (customer-supplied). UPS sizing specification provided: edge controller + network equipment ≤ 150 W; minimum 30-minute ride-through recommended. Optional Modbus UPS monitoring integration available.
GPS Antenna (for time sync)	PREFERRED	Optional — NTP (primary) and GPS antenna (secondary) time sync both supported. Achieves ≤ 1 ms accuracy with GPS; ≤ 10 ms with NTP. Time sync status monitored and alarmed on deviation > 10 ms.
CT/VT Transducers	PREFERRED	Excluded (customer-supplied). Analog input (4–20 mA / 0–10 V) integration for CT/VT-based measurements supported via commServer I/O mapping.
EMS Software License	REQUIRED	Included — SaaS subscription model. Full EMS optimization (essScheduler), price/load/yield forecasting, and degradation analytics included. Annual subscription per park; volume and multi-year discounts available.
SCADA Local Software	REQUIRED	Included — scadaServer deployed on edgeServer at each park. Browser-based React SCADA with one-line diagrams, real-time telemetry, alarm management, and local dispatch control. Accessible on-site via local LAN without internet.
SCADA Global / Fleet Management	REQUIRED	Included — Cloud-hosted aggregatorServer + VPP fleet dashboard supporting all 51 parks. Multi-group views (per client group), master O&M view for Lighthief, aggregator read-only access with scoped credentials. Kubernetes-hosted, multi-region.

HMI / Local Display	OPTIONAL	Optional — Browser-based PWA HMI accessible from any tablet or laptop on the site LAN. No dedicated touch-panel hardware required. Industrial touch-panel mounting option available on request.
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2.3 Detailed Bill of Materials

Please provide a detailed BoM for each site size category (Small / Medium / Large / XL), including part numbers, quantities, and unit prices.

3. PPC FUNCTIONALITY & GRID CODE COMPLIANCE

#	Question	Priority	Response
3.1	Is the PPC hardware-based (dedicated controller) or software running on the EMS platform?	REQUIRED	Software PPC running on edgeServer (NVIDIA Jetson Orin NX). Dedicated RTOS-priority scadaServer process with <100 ms control cycle. No separate hardware PPC controller required for any site size category.
3.2	Does the PPC support multi-PCS coordination (dispatching setpoints to up to 23 PCS units per park)?	REQUIRED	Yes — commServer dispatches active (P) and reactive (Q) setpoints simultaneously to up to 23 Kehua BCS-C-HUD PCS units per park via Modbus TCP. Individual per-unit control with aggregate park-level power target allocation.
3.3	Can the PPC provide grid-forming / Virtual Synchronous Generator (VSG) control modes?	PREFERRED	Planned Q3 2026 — VSG / grid-forming firmware module in development. Current production capability: droop-based grid-following. Grid-forming capability will be delivered as a software update; site-survey required for activation.
3.4	Is black-start capability supported?	PREFERRED	Planned Q4 2026 — Dependent on grid-forming capability (Q3 2026). Black-start sequence logic and islanding detection will be delivered as a software update. Site-survey and DSO agreement required per park.
3.5	Are droop control curves configurable (P-f and Q-V)? What range of droop settings (%) is supported?	REQUIRED	Yes — P-f and Q-V droop curves fully configurable via EMS portal. Default P-f droop: 4–6%; deadband: ± 10 mHz. Droop range: 1–20%. Q-V droop range: 1–10%. All settings changeable without service interruption.
3.6	Is ramp rate limiting configurable (kW/s or %/min)?	REQUIRED	Yes — Ramp rate limiting configurable in both kW/s and %/min. Applied at individual PCS level and at aggregate park level. Default: 10 kW/s per PCS. DSO-specific ramp limits stored per park profile and enforced automatically.
3.7	Can the PPC execute Volt-Var (Q(U)), Freq-Watt (P(f)), and Volt-Watt (P(V)) curves per EN 50549-2?	REQUIRED	Yes — All three EN 50549-2 response curves implemented: Q(U) Volt-Var, P(f) Freq-Watt, P(V) Volt-Watt. Configurable breakpoints for each curve. Activated autonomously by the edge controller

			without requiring cloud EMS communication.
3.8	What is the power factor control range (leading/lagging)?	REQUIRED	±0.9 lagging/leading to unity (1.0). Continuous PF adjustment per PCS unit. Fixed PF, PF(P) curve, and Q(U) modes all supported. PF targets dispatched via Modbus TCP to each Kehua PCS.
3.9	Does the PPC support LFSM-O (Limited Frequency Sensitive Mode — Over-frequency) with configurable activation threshold and droop?	REQUIRED	Yes — LFSM-O fully implemented. Configurable activation threshold (default 50.2 Hz), linear curtailment ramp to 50.7 Hz, and disconnect threshold. Executes autonomously at the edge independent of cloud EMS or DSO communication loss.
3.10	Can the PPC enforce DSO curtailment levels (100% / 60% / 30% / 0%) as discrete commands?	REQUIRED	Yes — DSO curtailment levels (100% / 60% / 30% / 0%) received as discrete IEC 60870-5-104 commands and applied within ≤500 ms. Curtailment overrides any active market dispatch schedule in the command priority hierarchy.

4. COMMUNICATION PROTOCOLS & DSO INTEGRATION

#	Question	Priority	Response
4.1	Is IEC 60870-5-104 supported natively or via a protocol gateway? (Native is mandatory for Cyprus DSO)	REQUIRED	Yes — Native. commServer implements a full IEC 60870-5-104 server stack on TCP/IP port 2404. No additional gateway hardware required. Tested with EAC-type DSO masters. Point list configurable per site.
4.2	Can you provide a sample IEC-104 point list for a BESS site? (Active power, reactive power, CB control, curtailment, telemetry, protection signals)	REQUIRED	Yes — Sample point list provided as Annex A (separate technical document). Includes: M_ME_NC_1 (active power MW), M_ME_NC_1 (reactive power Mvar), C_SE_NC_1 (active power setpoint), C_DC_NA_1 (CB open/close), M_SP_NA_1 (alarm/status), M_ME_NC_1 (frequency, voltage).
4.3	Is Modbus TCP/RTU supported for PCS and BMS integration? Can you interface with Kehua PCS (BCS-C-HUD) and Linyang BMS (BAMS) via Modbus TCP?	REQUIRED	Yes — Full Modbus TCP driver for Kehua BCS-C-HUD series confirmed (all register maps available). Linyang BAMS Level-3 Modbus TCP driver confirmed; BMU and BCMU data accessible via BAMS gateway. RS-485 RTU also supported.
4.4	Is IEC 61850 (MMS, GOOSE) supported for substation communication and protection relay integration?	PREFERRED	Yes — IEC 61850 MMS client/server and GOOSE subscription supported for MV switchgear and protection relay integration. Compatible with Schneider RMU and standard protection relay IEDs. SCL configuration import supported.
4.5	Is DNP3 supported for future-proofing?	OPTIONAL	Yes — DNP3 master stack available for future TSO or additional utility SCADA integration requirements.
4.6	What is the command acknowledgment time for IEC-104 commands? (Requirement: ≤ 3 seconds, target <500ms)	REQUIRED	Target <500 ms; maximum ≤ 3 s. ACT-CON acknowledgment sent immediately on command receipt; ACT-TERM after Modbus dispatch confirmation. Measured end-to-end in representative lab tests: median 180 ms, 99th percentile <450 ms on LTE link.
4.7	What is the power adjustment time from setpoint command to achieved output? (Requirement: ≤ 60 seconds, target <10 seconds)	REQUIRED	Target <10 s; maximum ≤ 60 s. Actual ramp time is determined by Kehua PCS ramp rate configuration (default 10 kW/s). At 10 MW park with 10 kW/s ramp, 0-100% transition completes in ~16 s. Sub-10 s achievable with higher ramp rate setting.

4.8	What control accuracy can be achieved relative to setpoint? (Requirement: $\pm 2\%$, target $\pm 1\%$)	REQUIRED	$\pm 1\%$ target; $\pm 2\%$ maximum. Closed-loop feedback from PCS active power measurement; setpoint corrected every 100 ms. Accuracy confirmed via grid-side metering cross-check. Derating conditions (temperature, SOC limits) flagged in SCADA when constraining accuracy.
4.9	What communication links are supported (4G/LTE, Ethernet point-to-point, fiber optic)? Does the system support automatic failover between communication paths?	REQUIRED	Yes — all links supported; automatic failover included. Dual-WAN: fiber/Ethernet primary + 4G/LTE backup. Failover detection ≤ 5 s; switchover transparent to IEC-104 DSO session (automatic reconnect). Fiber, Ethernet point-to-point, 4G, and satellite (optional) all supported.
4.10	Does the system provide REST API capability for cloud integration and third-party data exchange?	REQUIRED	Yes — Full REST API at <code>/api/v1</code> with OpenAPI/Swagger documentation. JWT Bearer token authentication; RBAC scopes per role. Supports telemetry ingestion, schedule management, ML inference, and historical data export. Rate limiting configurable per client.

5. BATTERY MONITORING — CELL-LEVEL MONITORING DEPTH

Our BESS uses a 3-level BMS with 96 BMUs per container. Each BMU provides 52 voltage samples and 12 temperature samples, yielding 4,992 voltage points and 1,152 temperature points per container.

#	Requirement	Priority	Response
5.1	Cell voltage monitoring: Can the EMS ingest and display all cell-level voltage data from the BMS (4,992 data points per container, sampled at $\leq 100\text{ms}$)?	REQUIRED	Yes — commServer ingests all $96 \text{ BMU} \times 52$ voltage channels from Linyang BAMS at $\leq 100 \text{ ms}$ polling cycle. All 4,992 voltage values stored in MongoDB telemetry_raw and visualised in SCADA cell-map view per container.
5.2	Cell temperature monitoring: Can the EMS ingest and display all cell-level temperature data (1,152 points per container, sampled at $\leq 200\text{ms}$)?	REQUIRED	Yes — $96 \text{ BMU} \times 12$ temperature channels at $\leq 200 \text{ ms}$. Displayed as colour-coded thermal map per container in SCADA. Per-BMU temperature trends logged historically for hotspot pattern analysis.
5.3	Internal resistance tracking: Does the EMS provide cell-level or module-level internal resistance tracking (AC impedance or DC resistance) as a derived metric from BMS data? At what resolution and frequency? Is this logged historically for degradation trending?	PREFERRED	Yes (module level). DC resistance derived from charge/discharge pulse response data; logged at 15-minute resolution. Module-level IR trends stored in MongoDB for long-term degradation analysis. Cell-level impedance spectroscopy planned as future enhancement.
5.4	Cell imbalance detection: Does the EMS monitor cell voltage imbalance and alert on threshold exceedance (e.g., $>30\text{mV}$ warning, $>50\text{mV}$ critical)? Is equalization status reported?	REQUIRED	Yes. Per-cell voltage deviation vs. container mean computed in real time. Configurable thresholds (default: $>30 \text{ mV} = \text{WARNING}$, $>50 \text{ mV} = \text{CRITICAL}$). Equalization status flag from BMS BAMS reported and displayed in SCADA. Alarm escalation rules apply.
5.5	SOC estimation accuracy: What SOC estimation accuracy can be guaranteed at the EMS level? Does the EMS apply enhanced algorithms (Kalman filtering, coulomb counting with drift correction, OCV correlation) on top of BMS-reported SOC? Can you provide a	REQUIRED	$\pm 1\%$ SOC accuracy guaranteed. Extended Kalman Filter (EKF) + coulomb counting with drift correction + OCV correlation fusion applied on top of BMS-reported SOC. Periodic full-charge OCV recalibration; SOC drift alarm if deviation $> 1\%$. Technical methodology brief available on request.

	technical brief on the SOC estimation methodology?		
5.6	SOH estimation with degradation model: Does the EMS include a real-time SOH estimation model that tracks actual degradation vs. OEM warranty curve? Is it physics-based (impedance growth + capacity fade) or empirical? Is it configurable per cell chemistry (LFP-specific)?	REQUIRED	Yes — physics-based model. Impedance growth + capacity fade model, parameterised specifically for LFP (EVE 314Ah cells). Actual SOH plotted against OEM warranty curve in real time. Configurable per cell chemistry. Degradation dashboard available at container/park/portfolio levels.

6. ADVANCED DEGRADATION ANALYTICS

#	Requirement	Priority	Response
6.1	Cycle counting algorithm: Does the EMS implement a rainflow cycle counting algorithm (or equivalent) to accurately count partial cycles and map them to equivalent full cycles? Is this per-container or per-cluster? Is the methodology aligned with IEC 62660-1?	REQUIRED	Yes. Rainflow cycle counting algorithm per container, aligned with IEC 62660-1 methodology. Partial cycles mapped to equivalent full cycles (EFC) using depth-weighted counting. Per-container and per-cluster aggregation both available. EFC count displayed in degradation dashboard.
6.2	Throughput-based degradation tracking: Does the EMS track cumulative energy throughput (MWh in + MWh out) per container/cluster and correlate it with capacity degradation? Is this presented as a degradation-per-MWh metric?	REQUIRED	Yes. Cumulative MWh-in and MWh-out tracked per container from commissioning. Degradation-per-MWh metric (% SOH loss per 1,000 MWh) displayed in degradation dashboard. Correlated with measured capacity fade from periodic capacity tests.
6.3	Calendar aging model: Does the EMS include a calendar aging model that accounts for time-at-temperature and SOC-at-rest effects? Is this separable from cycle aging in the degradation dashboard?	PREFERRED	Yes. Calendar aging model includes time-at-temperature and SOC-at-rest aging terms. Shown as a separate component in the degradation dashboard, clearly distinguished from cycle-induced aging. LFP calendar aging coefficients configurable per OEM specification.
6.4	Dynamic DoD impact modelling: Does the EMS dynamically model the impact of varying Depth of Discharge on battery degradation? Can it adjust dispatch strategies to optimize between revenue and battery lifetime? Is the DoD-cycle life curve configurable?	PREFERRED	Yes. Configurable DoD-cycle-life curve (LFP default provided; fully customisable). Optimizer dynamically adjusts DoD based on marginal revenue vs. marginal degradation cost. Asset owner can set maximum acceptable degradation cost (€/MWh) to constrain dispatch depth.
6.5	Predictive EOL forecasting: Does the EMS provide a predictive End-of-Life forecast based on actual usage data (cycles,	PREFERRED	Yes. EOL date projection updated daily based on cumulative cycles, throughput, temperature history, and calendar time. Flags when projected SOH will breach the 70% warranty threshold. Monte Carlo

throughput, temperature history, calendar time)? Can it project the date at which SOH will reach the warranty threshold (70%)? Is this updated dynamically?

uncertainty range (P10/P50/P90) provided. Used to generate warranty claim reports.

7. THERMAL & SAFETY MONITORING

#	Requirement	Priority	Response
7.1	Thermal runaway early detection: Does the EMS integrate BMS temperature data, gas detection (H ₂ , CO), and smoke detection signals to provide multi-layer thermal runaway early warning? What is the detection methodology?	REQUIRED	Yes — multi-layer detection. BMS temperature rate-of-rise analysis + H ₂ combustible gas concentration + CO concentration + smoke detector status fused into an anomaly score. Alarm generated before BMS hardware protection activates. Methodology: threshold on rate-of-rise >2°C/min on any BMU combined with any gas concentration exceedance.
7.2	Hotspot localization: Does the EMS provide a thermal map or hotspot visualization feature that leverages per-BMU temperature data? Can it identify localized thermal anomalies (e.g., specific rack running consistently hotter)?	PREFERRED	Yes. Per-BMU thermal map rendered as colour-coded heat map in SCADA. Racks and modules with temperature >ΔT above container mean are highlighted. Trend view shows developing hotspots over time. Alarm triggered when specific BMU consistently exceeds configurable ΔT from mean.
7.3	HVAC/cooling performance monitoring: Does the EMS monitor cooling system metrics (COP, coolant flow rate, inlet/outlet delta-T, compressor run-time, energy consumption)? Are HVAC efficiency degradation trends tracked?	PREFERRED	Yes. Cooling system COP, inlet/outlet delta-T, compressor run-time, and auxiliary energy consumption tracked as real-time KPIs via Modbus or analog I/O. HVAC efficiency degradation trended over time. Alarm when delta-T exceeds setpoint or COP drops below threshold.
7.4	Fire system status integration: Does the EMS integrate with the fire detection and suppression system? Can it display fire alarm zone status, suppression system readiness, and actuate emergency shutdown commands?	REQUIRED	Yes. FACP (Fire Alarm Control Panel) integrated via Modbus TCP or dry-contact I/O. Fire alarm zone status, suppression system readiness (pressurised/discharged), and manual/automatic mode displayed in SCADA. Emergency shutdown command (EMS → PCS trip) actuated automatically on confirmed fire trip signal.
7.5	Gas detection integration: Does the EMS receive and process H ₂ combustible gas and CO sensor signals? Are gas concentration trends logged and alarmed at configurable thresholds?	REQUIRED	Yes. Analog (4–20 mA) or Modbus gas sensor inputs for H ₂ and CO. Configurable alarm thresholds (warning and critical levels). Gas concentration trends logged at 1-second resolution. Alarm escalation rules apply; CRITICAL gas alarm triggers automatic power reduction response.

8. FAULT DETECTION & DIAGNOSTICS

#	Requirement	Priority	Response
8.1	Real-time fault classification: Does the EMS provide fault classification beyond BMS-level alarms? (a) Automatic categorization by type/severity/location, (b) correlation of multiple alarm signals to identify root fault vs. cascading effects, (c) fault history analytics with trend analysis.	REQUIRED	Yes. (a) Faults categorised by type (thermal/electrical/communication/protection), severity (INFO/WARNING/CRITICAL/EMERGENCY), and location (container/PCS/BMS/grid). (b) Cross-layer correlation of BMS + PCS + thermal + grid signals to distinguish root fault from cascading effects. (c) Fault history with trend analytics; full context snapshot (SOC, temperature, power flow) stored per event.
8.2	Predictive maintenance analytics: Does the EMS include predictive maintenance features? (a) Anomaly detection using statistical/ML models on sensor data trends, (b) remaining useful life estimation for key components (cells, contactors, cooling system), (c) automated maintenance scheduling based on condition monitoring. What data history is required before predictive models become effective?	PREFERRED	Yes (partial). (a) Anomaly detection using LSTM autoencoder + isolation forest on cell, PCS, and cooling sensor trends — in production. (b) RUL estimation for cells and cooling components — in beta. (c) Condition-based maintenance scheduling triggers: configurable threshold-based alerts — in production. Minimum 90 days of baseline data required before ML models reach full effectiveness.
8.3	Root cause analysis engine: Does the EMS provide automated or assisted root cause analysis (RCA) for fault events? Can it correlate time-stamped data across BMS, PCS, thermal, and grid signals to help operators identify the origin of a fault chain?	PREFERRED	Yes — assisted RCA. For each fault event, a time-correlated event timeline is automatically constructed across BMS/PCS/thermal/grid signals. The timeline identifies the first-triggering signal vs. subsequent cascading effects. Operator-guided workflow in EMS portal allows annotating findings; RCA reports exportable as PDF.
8.4	Alarm prioritization & management: Does the EMS provide: (a) severity-based filtering (info/warning/critical/emergency), (b) alarm suppression for known conditions, (c) alarm shelving during maintenance, (d) escalation rules (e.g., unacknowledged critical → SMS/email after X minutes)?	REQUIRED	Yes — all four. (a) INFO/WARNING/CRITICAL/EMERGENCY severity levels with independent notification channels. (b) Configurable alarm suppression rules for expected conditions. (c) Alarm shelving with mandatory timeout and visibility in dedicated shelved-alarm queue. (d) Configurable escalation: unacknowledged CRITICAL → SMS/email to on-call engineer after configurable timeout (default 10 min); EMERGENCY: immediate multi-channel notification.

9. DATA INFRASTRUCTURE & LOGGING

#	Requirement	Priority	Response
9.1	High-resolution data logging: What is the minimum data logging resolution? Can the EMS log at 1-second resolution for real-time measurements? Is sub-second resolution (100ms, 200ms) available for specific signals?	REQUIRED	Yes. Standard: 1-second resolution for all power, voltage, current, and frequency measurements. Sub-second: 100 ms for cell-level BMS critical signals (voltage, temperature per BMU); 200 ms for gas sensor readings. Resolution configurable per signal class.
9.2	Event-triggered recording: Does the EMS support event-triggered high-resolution recording (increase logging frequency when a fault/transient is detected)? Similar to a disturbance recorder or oscillography function? What triggers are configurable?	PREFERRED	Yes. Disturbance recorder mode: on fault/transient trigger, all relevant signals automatically elevated to maximum resolution. Configurable trigger list includes: overcurrent, voltage transient, frequency excursion (>50.2 Hz / <49.8 Hz), BMS CRITICAL alarm, gas concentration threshold, PCS fault code. Pre-trigger buffer: 30 seconds; post-trigger capture: 120 seconds (configurable).
9.3	Time synchronization: Does the EMS support NTP and/or GPS-based time synchronization? What time accuracy is achieved? Is time sync status monitored and alarmed?	REQUIRED	Yes — NTP (primary) and GPS antenna (secondary). ≤1 ms accuracy with GPS; ≤10 ms with NTP stratum-2. Time sync status monitored continuously; CRITICAL alarm if deviation exceeds 10 ms or sync source is lost. GPS PPS input supported on Jetson Orin NX.
9.4	Data retention policy: What are the data retention periods for: (a) raw real-time measurements, (b) event logs, (c) performance data, (d) alarm history? Are these configurable?	REQUIRED	Yes — all configurable. Defaults: (a) Raw 1-second measurements: 90 days edge / 1 year cloud; (b) Event/disturbance recordings: 5 years; (c) 15-minute performance aggregates: 10 years; (d) Alarm history: 10 years; (e) Degradation/SOH history: asset lifetime; (f) Market bid archive: 7 years. All retention periods adjustable via EMS administration portal.
9.5	API & data export: Is a REST API available for external data access? What export formats are supported (CSV, JSON, IEC 61968 CIM, Excel)? Is there a rate limit on API calls?	REQUIRED	Yes. REST API at /api/v1 with OpenAPI/Swagger documentation. Supported export formats: CSV, JSON, IEC 61968 CIM, Excel (XLSX). Rate limits configurable per API key (default 1,000 req/min for aggregation queries; 10,000 req/min for telemetry streams). Bulk historical export endpoint available.
9.6	Data ownership: Who owns the operational data generated by the EMS? Can	REQUIRED	Asset owner owns all data in perpetuity. Full historical data accessible at any time via API and bulk export. On contract termination, complete data export

	the client/asset owner access all historical data in perpetuity? What happens to stored data if the EMS service contract is not renewed?		package (all formats) delivered within 30 days. Data not retained by diffusedEnergy beyond a 90-day grace period after contract end. No data held hostage.
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10. EMS OPTIMIZATION CAPABILITY

#	Requirement	Priority	Response
10.1	Multi-objective optimization engine: Describe the optimization engine architecture: (a) Is it a multi-objective optimizer (maximize revenue while minimizing degradation and maintaining grid obligations)? (b) What optimization algorithm is used (LP, MILP, dynamic programming, heuristic)? (c) What is the optimization horizon (rolling 24h, 48h, week-ahead)? (d) What is the dispatch resolution (1 min, 5 min, 15 min)?	REQUIRED	Yes. (a) Multi-objective: maximise revenue subject to minimise degradation cost and fulfil grid obligations. (b) MILP (Mixed Integer Linear Programming). (c) Rolling 24h horizon re-optimised every 15 minutes; 48h lookahead option available. (d) 15-minute dispatch resolution standard; 5-minute in development. Computation time: <30 seconds per run on cloud, <2 minutes on edge fallback.
10.2	Revenue stacking optimization: Can the EMS simultaneously optimize across multiple revenue streams (e.g., energy arbitrage + FCR + aFRR + curtailment avoidance)? How does it handle conflicting obligations between services? Can revenue stream priorities be configured?	REQUIRED	Yes. Simultaneous optimisation across energy arbitrage, FCR, aFRR, and curtailment avoidance. Conflict resolution: FCR/aFRR obligations treated as hard constraints; arbitrage as soft objective. Revenue stream priority weighting fully configurable per asset owner preferences. Committed ancillary service capacity reserved before arbitrage allocation.
10.3	Constraint-aware dispatch: Does the optimizer account for: (a) SOC limits (min/max), (b) ramp rate limits, (c) C-rate limits, (d) temperature-dependent power derating, (e) grid connection capacity limits, (f) contractual availability requirements?	REQUIRED	Yes — all six constraints. (a) SOC min/max per site profile; (b) ramp rate per PCS and park aggregate; (c) C-rate limits per container; (d) temperature-dependent derating from live BMS thermal data; (e) grid connection capacity per DSO agreement; (f) contractual availability window enforced as hard constraint. All parameters configurable per park.
10.4	Degradation-aware dispatch: Does the optimization engine incorporate a degradation cost function? (a) Does it assign a €/MWh degradation cost to each	REQUIRED	Yes — all four. (a) €/MWh degradation cost assigned per dispatch cycle. (b) Cycling reduced when marginal revenue < marginal degradation cost; configurable floor bid price. (c) LFP-specific DoD–cycle-life curve configurable per container OEM spec. (d) DoD adjusted dynamically: deep cycling only when high price spread justifies degradation cost.

	charge/discharge cycle? (b) Can it reduce cycling when marginal revenue is below marginal degradation cost? (c) Is the degradation model configurable per LFP cycle life curves? (d) Can it adjust DoD dynamically based on price signals?		
10.5	Stochastic optimization: Does the optimizer use stochastic methods to account for uncertainty in price forecasts, RES generation, and load? (a) Is it scenario-based (Monte Carlo) or robust optimization? (b) How many scenarios are evaluated? (c) What is the computation time per optimization run?	PREFERRED	Planned Q4 2026. Current: deterministic with confidence-weighted forecast inputs. Planned: scenario-based Monte Carlo (50–200 scenarios per run). Current deterministic computation: <30 s cloud / <2 min edge. Monte Carlo target: <5 min on edge hardware. Robust optimisation variant under evaluation.
10.6	Degradation-aware bidding logic: Does the EMS incorporate battery degradation costs into market bidding decisions? (a) Can it set a minimum bid price that covers the degradation cost of the dispatch? (b) Can it refuse to bid into low-value markets when degradation cost exceeds revenue? (c) Is this configurable per asset owner's risk preferences?	PREFERRED	Yes. (a) Minimum bid price floor = modelled degradation cost + variable O&M; configurable. (b) Bids below floor price automatically withheld. (c) All parameters configurable per asset owner; risk tolerance setting adjusts aggressiveness of cycling vs. asset preservation trade-off.
10.7	Peak shaving / load shifting: Does the EMS support peak shaving and load shifting optimization? Can these modes operate alongside market participation?	REQUIRED	Yes. Dedicated peak shaving and load shifting modes implemented. Both can operate concurrently with market participation; configurable mode priority determines how conflicting objectives are resolved. Site-specific peak demand thresholds and tariff structures configurable.

10.8	Arbitrage optimization: Does the EMS include a time-of-use arbitrage module? What inputs does it use (price forecasts, historical prices, price signals)?	REQUIRED	Yes. Dedicated time-of-use arbitrage module in energyPrice + essScheduler services. Inputs: day-ahead price forecasts (ENTSO-E / market API), historical price curves, intraday price signals (when available), volatility bands. Optimizer identifies charge windows (low price) and discharge windows (high price) subject to SOC and degradation constraints.
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11. MARKET INTEGRATION & REVENUE OPTIMISATION

#	Requirement	Priority	Response
11.1	Day-ahead market interface: Does the EMS provide a direct interface to day-ahead electricity markets? (a) What market platforms are supported? (b) Can it generate and submit day-ahead bid schedules automatically? (c) Is there a day-ahead optimization module producing optimal charge/discharge schedules based on price forecasts? (d) How does it interface with third-party aggregators?	REQUIRED	Yes. (a) Aggregator API integration (REST/JSON); direct exchange connectivity planned post-Cyprus market rule publication. (b) Day-ahead bid schedules generated automatically from price forecasts; API-pushed to aggregator platform. (c) Day-ahead optimisation module in essScheduler produces optimal 24h charge/discharge schedule. (d) Aggregator integration via REST API with scoped JWT token; read/write access to dispatch schedules.
11.2	Intraday continuous trading: Does the EMS support intraday continuous trading? (a) Can it revise dispatch schedules in near-real-time based on updated price signals? (b) What is the minimum lead time for schedule revision? (c) Is this relevant to the Cyprus market structure (currently under development by CERA/TSO)?	PREFERRED	Yes (partial). (a) Schedule revision supported on every 15-minute optimisation cycle based on updated price signals. (b) Minimum lead time: 15 minutes. (c) Cyprus intraday market structure monitored; functionality will be updated to match CERA/TSOC rules when published. Direct intraday exchange connectivity planned for H2 2026.
11.3	Ancillary services (FCR/aFRR): Does the EMS support bidding and delivery for frequency regulation services? (a) Can it calculate available capacity for bidding (accounting for SOC, degradation, other commitments)? (b) What is the activation time for FCR (target <1 second) and aFRR (target <30 seconds)? (c) Can FCR droop settings (4–6%) and deadband (± 10 mHz) be configured? (d) Is the response accuracy within $\pm 2\%$ of setpoint?	REQUIRED	Yes. (a) Available bidding capacity calculated in real time accounting for SOC, degradation state, temperature derating, and other committed obligations. (b) FCR activation <1 s; aFRR activation <30 s. (c) FCR droop 4–6% and deadband ± 10 mHz fully configurable. (d) Response accuracy $\pm 1\%$ target; $\pm 2\%$ maximum — confirmed by closed-loop control.

11.4	Automatic bid submission: Can the EMS automatically submit bids to market/TSO platforms without manual intervention? (a) What market platforms are currently integrated? (b) Is the Cyprus market (CERA/TSO) currently supported or planned? (c) What is the failsafe if automatic submission fails?	PREFERRED	Partial. (a) Automatic schedule generation confirmed; push to aggregator API (REST) automated. Direct exchange integration: to be added per market. (b) Cyprus TSOC/CERA integration planned once market API specifications are published. (c) Failsafe on submission failure: hold last committed schedule + immediate operator notification + email alert to named contact.
11.5	Latency risk in ancillary markets: How does the EMS handle latency risk for ancillary service delivery? (a) Is communication latency monitored in real-time? (b) Does the system pre-position (anticipatory dispatch) to compensate for communication delay? (c) Is there an alarm when measured latency exceeds the threshold for the active service type?	PREFERRED	Yes. (a) RTT latency to TSO/aggregator monitored every 5 seconds; logged as time-series KPI. (b) Anticipatory pre-positioning activated when latency exceeds 80% of the service activation time threshold. (c) WARNING alarm when latency exceeds configurable threshold; CRITICAL alarm if latency prevents SLA-compliant delivery; active service de-commitment offered if sustained.

12. FORECAST INTEGRATION

#	Requirement	Priority	Response
12.1	Price forecasting module: Does the EMS include a built-in electricity price forecasting module? (a) What data sources does it use? (b) What forecast horizons are supported (day-ahead, week-ahead)? (c) What is the forecast accuracy (MAPE)? (d) Is it pre-trained for the Cyprus market or does it require market-specific calibration?	REQUIRED	Yes. Built-in energyPrice service (Port 7001). (a) Data sources: ENTSO-E Transparency Platform, market price APIs, local DSO signals. (b) Day-ahead and week-ahead horizons supported. (c) Typical MAPE 5–10% day-ahead for calibrated models; heavily dependent on market data availability. (d) Cyprus market requires calibration; training data request: minimum 12 months of historical prices; Cyprus-specific model built during commissioning phase.
12.2	Volatility modelling: Does the EMS model price volatility for dispatch optimization? (a) Can it adjust bidding strategy based on expected price spreads? (b) Does it account for seasonal and intraday volatility patterns? (c) Can it identify high-value arbitrage windows dynamically?	PREFERRED	Yes. (a) Price volatility band (σ) computed alongside point forecast; optimizer adjusts bidding aggressiveness based on expected price spread. (b) Seasonal and intraday patterns modelled using historical decomposition. (c) High-value arbitrage windows identified dynamically by comparing expected discharge revenue against charge cost + degradation cost floor.
12.3	Load & RES forecast integration: Can the EMS ingest external load and renewable energy (solar/wind) generation forecasts? (a) What input formats are supported (API, CSV, MQTT)? (b) Can it integrate with third-party forecast providers? (c) Is there a built-in solar generation forecast? (d) How does the RES forecast influence dispatch decisions?	PREFERRED	Yes. (a) REST API, CSV upload, and MQTT all supported for forecast ingestion. (b) Third-party forecast provider API integration supported via configurable connector. (c) Built-in solar/wind generation forecast via yieldProfile service (Earth-2 Studio + OpenMeteo fusion). (d) RES forecast fed directly into essScheduler: reduces charge during high-generation windows; schedules discharge when RES output is low.
12.4	Scenario-based planning: Does the EMS support scenario-based planning for dispatch strategy evaluation? (a) Can	PREFERRED	Partial — UI planned Q3 2026. Backend scenario engine implemented; operators can currently compare strategies via API. EMS portal UI for side-by-side what-if comparison (multiple price/RES forecast scenarios)

	operators compare multiple dispatch strategies before committing? (b) Are "what-if" simulations available? (c) Can scenarios include different price forecast assumptions?		under development for Q3 2026 release. Scenario export as PDF/CSV included.
12.5	Machine learning forecasting engine: Does the EMS use machine learning for any forecasting function? (a) What ML techniques are employed (LSTM, gradient boosting, ensemble)? (b) Is the model re-trained automatically as new data is collected? (c) What minimum data history is required? (d) Can model performance be audited by the client?	OPTIONAL	Yes. (a) Hugging Face Chronos-2 (120M parameter time-series transformer), LSTM ensembles, gradient boosting (XGBoost/LightGBM) all in use across load, price, and yield forecast pipelines. (b) Automatic weekly retraining on new data; MLflow tracks all model versions. (c) Minimum 90 days for effectiveness; 12 months for full seasonal model. (d) MLflow dashboard with per-metric performance history accessible to client administrators.

13. REMOTE DISPATCH & FAIL-SAFE LOGIC

#	Requirement	Priority	Response
13.1	Manual override hierarchy: What is the command priority hierarchy? (Required: Protection > DSO SCADA curtailment > DSO CB control > Market dispatch. Local protection must be non-overridable.)	REQUIRED	Implemented. Priority hierarchy (highest first): 1. Hardware protection relays (non-overridable firmware level). 2. DSO SCADA curtailment via IEC 60870-5-104. 3. DSO CB control. 4. Local EMS operator. 5. Market dispatch / aggregator schedule. Local hardware protection cannot be disabled by any software layer.
13.2	Real-time dispatch override: Can DSO curtailment commands override market dispatch in real-time? What is the response time?	REQUIRED	Yes. DSO curtailment via IEC-104 ACTCON immediately overrides market dispatch at the edge controller. Override response time: <100 ms from IEC-104 command receipt to PCS setpoint change. Market schedule is suspended and logged as "DSO-curtailed" event; automated reconciliation submitted to aggregator post-event.
13.3	Fail-safe fallback logic: What happens when EMS communication is lost? (a) Does the edge controller revert to a pre-programmed safe mode? (b) What is the communication loss detection timeout? (c) Can a fallback dispatch schedule be pre-loaded? (d) Is there "last known good" state retention? (e) How is communication restoration handled?	REQUIRED	Fully implemented. (a) Yes — edge controller enters pre-programmed safe mode (configurable: idle, fixed SOC hold, or last schedule). (b) Communication loss detection: configurable timeout (default 30 s). (c) 24-hour fallback schedule pre-loadable and auto-executed on comms loss. (d) Last-known-good state retained in non-volatile memory. (e) On restoration: edge controller resyncs state, downloads updated schedule, and resumes full control within 60 s; reconciliation log uploaded automatically.
13.4	Autonomous frequency/voltage response: Can the edge controller execute automatic frequency-based (LFSM-O: curtailment 50.2–50.7Hz, charging at 50.7Hz) and voltage-based (P(V): 1.07–1.10 pu) protection responses independently of EMS communication?	REQUIRED	Yes, fully autonomous. LFSM-O (f: 50.2–50.7 Hz curtailment ramp; hard stop & inhibit charge at 50.7 Hz) and P(V) (1.07–1.10 pu linear curtailment) operate inside the PPC firmware independently of cloud/central EMS connection. Parameters configurable remotely and stored locally. Active even during total backhaul loss.
13.5	Communication failure scenarios: Describe the handling of: (a) EMS-to-site loss, (b) site-to-DSO loss	REQUIRED	All scenarios handled. (a) EMS→site loss: edge controller executes fallback schedule; alarm raised; 4G backup activates (timeout 30 s, configurable). (b) Site→DSO loss: IEC-104 auto-reconnect with

	(IEC-104 reconnection), (c) BMS-to-EMS loss, (d) multi-path simultaneous failure. What are the configurable timeouts at each level?		configurable T1/T2/T3 timers; last DSO setpoint maintained during gap; STARTDT on reconnection. (c) BMS→EMS loss: affected string isolated from dispatch; safety-hold on PCS; alarm escalated. (d) Multi-path failure: hardware protection relays remain active; site held at last safe state; on-call engineer notified within 2 minutes.
13.6	Edge controller autonomy: Can the edge controller execute a pre-loaded dispatch schedule autonomously if central EMS connection is lost? (a) How long can it operate autonomously? (b) Can the schedule be updated remotely before disconnection? (c) Does it log operations during autonomous mode for later sync?	PREFERRED	Yes. (a) Autonomous operation: up to 7 days from pre-loaded schedule; UPS-backed local hardware sustains controls. (b) Schedule update: remotely pushed up to the point of communication loss; next-cycle schedule always pre-staged. (c) All operations during autonomous mode logged locally at 1-second resolution; full log sync on reconnection with timestamp reconciliation.

14. REDUNDANCY & HIGH AVAILABILITY

#	Requirement	Priority	Response
14.1	EMS server redundancy: What is the EMS server architecture for redundancy? (a) Dual-server (active/standby) supported? (b) Failover time? (c) Is data synchronized between primary and standby?	PREFERRED	Yes. (a) Cloud-deployed Kubernetes cluster with automatic pod restart and horizontal scaling; on-premises: dual-server active/standby with HAProxy available. (b) Cloud failover <60 s; on-premises active/standby <30 s with proper infrastructure. (c) Time-series data replicated via InfluxDB clustering or Prometheus remote-write; PostgreSQL replication (streaming replication).
14.2	Redundant communication lines: Does the EMS support automatic failover between communication paths (e.g., fiber primary, 4G backup)? What is failover detection time? Is switchover seamless?	REQUIRED	Yes. Standard deployment: fiber (primary) + 4G/LTE (secondary) with dual-SIM router (Teltonika RUT956 or equivalent). Failover detection: ICMP keepalive; failover trigger: 3 missed keepalives (configurable, typically 15 s). Switchover is seamless — IEC-104 reconnects automatically; VPN tunnel re-established on backup link within 30 s.
14.3	UPS requirements: What are the UPS requirements for the EMS edge controller and network equipment? What minimum runtime is needed (ride-through vs. graceful shutdown)? What is the power consumption?	REQUIRED	UPS included in supply. Edge controller + network equipment: offline/line-interactive UPS, minimum 15-minute ride-through at full load. Power consumption: PPC/RTU + router + switch + controls $\approx$ 200–400 W (site-size dependent). Graceful shutdown sequence initiated at 5-minute remaining capacity. Recommended: APC Smart-UPS 1500VA or equivalent.
14.4	Hot-standby failover: Is hot-standby failover supported? (a) Edge controller level: dual-PLC? (b) SCADA Local: active/standby? (c) SCADA Global: automatic failover? (d) What is the RTO (Recovery Time Objective) at each level?	PREFERRED	Partial. (a) Dual-PLC edge controller: available as option for XL sites (>15 MW), RTO <5 s. (b) SCADA Local: active/standby supported, RTO <60 s. (c) SCADA Global: Kubernetes rolling failover, RTO <60 s cloud / <30 s on-premises. (d) Summary: Edge <5 s (dual) / <30 s (single restart); SCADA Local <60 s; SCADA Global <60 s.
14.5	System availability SLA: What system availability can be guaranteed? (Target: $\geq 99.5\%$ for SCADA, $\geq 99.9\%$ for edge controller)	REQUIRED	Committed SLA. SCADA Local/Global: $\geq 99.5\%$ annual availability. Edge controller (hardware-dependent): designed for $\geq 99.9\%$ with UPS and 4G backup; hardware MTBF >50,000 h. Planned maintenance windows excluded from availability calculation with 5-day advance notice. SLA breach credits specified in MSA.

15. CYBERSECURITY & NIS2 COMPLIANCE

#	Requirement	Priority	Response
15.1	Role-based access control (RBAC): Is RBAC implemented? What roles are supported (administrator, operator, viewer, aggregator)? Is multi-factor authentication (MFA) supported?	REQUIRED	Yes, fully implemented. Roles: System Admin (full), O&M Operator (control + acknowledge alarms), Read-Only Viewer, Aggregator (scoped read + schedule push), Auditor (log read-only). MFA: TOTP (Google Authenticator / Authy) enforced for Admin and Operator roles; optional for Viewer. JWT tokens with configurable 1–8 hour expiry. LDAP/AD integration available. Keycloak implemented for multi-domain multi-level user management.
15.2	Network segmentation: Does the architecture support OT/IT network separation with defined security zones? Is an IEC 62443 certified industrial firewall included or recommended?	REQUIRED	Yes. Standard architecture: VLAN-based OT/IT separation with DMZ zone. OT zone: BMS/PCS Modbus RTU, IEC-104 to DSO. IT zone: SCADA Local, HMI, EMS edge. DMZ: router, firewall, VPN concentrator. Recommended industrial firewall: Fortinet FortiGate-60F or Cisco IE3400 (IEC 62443 SL2 certified). FW rule templates provided in commissioning package.
15.3	Encrypted communication: What encryption standards are used? (Requirements: TLS 1.3 minimum for external, TLS 1.2+ for SCADA, encrypted tunnel for Modbus TCP, IPSec/OpenVPN for remote access)	REQUIRED	Fully compliant. External APIs: TLS 1.3 (AES-256-GCM). SCADA Local–Global: TLS 1.2+. Modbus TCP: encapsulated in WireGuard VPN tunnel (ChaCha20 or proprietary XIOTIC post-quantum algorithm (available add-on license)). Remote access: WireGuard (primary) or OpenVPN (fallback) with certificate-based authentication. IEC-104: option for TLS transport (IEC 60870-5-7). All TLS certificates rotated annually; Let's Encrypt or self-signed with custom CA.
15.4	Intrusion detection system: Does the EMS or network architecture include IDS/IPS? (a) Network-based or host-based? (b) Does it monitor for OT-specific threats (Modbus protocol anomalies, unauthorized SCADA commands)? (c) Is it integrated with the alarm system?	PREFERRED	Partial — HIDS included; network IDS optional. Nvidia Morpheus available (optional add-on). (a) Host-based IDS (OSSEC/Wazuh) deployed on SCADA servers. Network IDS: available as optional add-on (Zeek/Suricata on managed switch). (b) OT-specific rules: Modbus function code anomaly detection, rogue ASDU type monitoring on IEC-104. (c) IDS events injected into EMS alarm bus — appear alongside process alarms in dashboard; email/SMS notification.
15.5	Secure firmware/software update: What is the secure update process? (a) Cryptographically signed updates? (b) Rollback supported? (c) Staged deployment before fleet-	REQUIRED	Yes, secure CI/CD pipeline. (a) All update packages signed with GPG; signature verified before installation. (b) One-click rollback to previous version; last 3 versions retained. (c) Staged rollout: 1 canary site → 5 sites → full fleet (each stage 48 h observation). (d) Feature releases quarterly; patch releases as needed. (e) Critical security patches: 72-

	wide rollout? (d) Update frequency? (e) Security patch SLA?		hour SLA; high severity: 7 days; medium/low: next quarterly release.
15.6	Audit trail logging: Are all user actions and commands logged with timestamps? What is the minimum retention period? Is tamper-evident logging supported?	REQUIRED	Yes. All user logins, command submissions, setpoint changes, alarm acknowledgements, and API calls logged with UTC timestamp, user ID, and source IP. Retention: minimum 5 years (configurable up to 10 years). Tamper-evident: logs forwarded in real time to append-only remote syslog (WORM policy); hash chain on local log files (SHA-256 per day). Audit log export to CSV/PDF on demand.
15.7	Penetration testing: Has the EMS platform undergone independent penetration testing? (a) By which firm and when? (b) Can results be shared? (b) Is periodic pen testing included in the maintenance contract? (d) Does the company hold ISO 27001, IEC 62443 SL, or other cybersecurity certifications?	PREFERRED	Planned. Internal security review completed; independent third-party penetration test scheduled for Q2 2026. Results summary (excluding critical findings) shareable under NDA. Annual pen test available as contract add-on. Current certifications: in process for ISO 27001 (target Q3 2026). IEC 62443 SL1 self-assessment completed; SL2 certification assessment planned H1 2027.
15.8	NIS2 Directive compliance: Is the platform fully compliant with the EU NIS2 Directive for critical infrastructure (energy sector)? Provide a compliance statement or gap analysis.	REQUIRED	Aligned; formal statement in Q3 2026. Architecture designed to NIS2 Article 21 requirements: multi-layer access control, encrypted communications, incident response procedures, supply chain security assessments, and business continuity measures. NIS2 gap analysis conducted internally (2025-Q4); no blocking gaps identified. Formal compliance statement with supporting evidence to be issued Q3 2026 as part of ISO 27001 certification process.

16. KPI MONITORING & REPORTING

#	KPI	Priority	Response
16.1	Availability (%): Does the EMS track and report system availability? (Target: $\geq 97\%$)	REQUIRED	Yes. System availability calculated automatically per site: $\text{uptime} / (\text{uptime} + \text{unplanned downtime}) \times 100$. Dashboard shows real-time availability %, 30-day rolling average, and YTD. Monthly report includes availability against $\geq 97\%$ target with breakdown of downtime cause codes (comms, hardware, software, planned maintenance).
16.2	Round-trip efficiency (%): Does the EMS calculate and display AC-AC round-trip efficiency including PCS losses, auxiliaries, and thermal management? (System specification: $\geq 86\%$)	REQUIRED	Yes. AC-AC RTE calculated as: $E_{\text{discharge_AC}} / E_{\text{charge_AC}}$, including PCS conversion losses, BMS parasitic, HVAC, and auxiliary loads. Displayed per container, per park, and portfolio. Efficiency trend chart with rolling 30-day average; alerting when RTE drops $> 2\%$ below baseline (indicating degradation or auxiliary issue). Compared against $\geq 86\%$ specification benchmark.
16.3	Revenue per MW: Does the EMS calculate revenue per MW? Can it break down by source (arbitrage, FCR, aFRR, curtailment avoidance)? Available at park and portfolio level?	REQUIRED	Yes. Revenue tracked per kWh and per kW by service: arbitrage, FCR, aFRR, curtailment avoidance, and other ancillary. Visible at container, park, group, and portfolio levels on the SCADA Global dashboard. Monthly performance report exports revenue breakdown as table and chart. Actuals vs. budget comparison view available.
16.4	Degradation cost per MWh: Does the EMS calculate degradation cost per MWh dispatched? What cost model is used? Is this factored into dispatch optimization and displayed as a KPI?	PREFERRED	Yes. Degradation cost model: $\Delta\text{SOH per cycle} \times \text{replacement cost per kWh}$. Model parameters: user-configurable BESS replacement cost (€/kWh), cycle stress function (DoD, rate, temperature). KPI displayed as €/MWh dispatched, updated daily. Cost floor enforced in dispatch optimizer — dispatch rejected if revenue $<$ degradation cost floor (configurable bypass possible).
16.5	Response time compliance: Does the EMS track DSO command response times and report compliance rates against regulatory requirements?	REQUIRED	Yes. Every DSO SCADA command (receive timestamp $\rightarrow$ PCS ramp start) logged with ms precision. Compliance KPI: % commands with < 100 ms response; reported monthly. Alarm raised if any command exceeds 200 ms. Quarterly DSO compliance report auto-generated with command-level detail exportable for regulatory submission.
16.6	SOC accuracy drift: Does the EMS monitor SOC estimation accuracy over time? Is there periodic SOC recalibration (e.g., full	PREFERRED	Yes. SOC estimation accuracy tracked by comparing estimated vs. measured capacity on each calibration cycle. Calibration trigger: automatic every 30 days (configurable) via full charge to OCV reference. SOC drift alarm: WARNING at $> 3\%$ drift vs. last calibration; CRITICAL at $> 5\%$. Forced recalibration can be

	charge to known OCV)? Can SOC drift be alarmed?		initiated by operator or triggered automatically upon drift alarm.
16.7	Auxiliary consumption ratio: Is auxiliary consumption tracked as a real-time KPI? Can it be broken down by subsystem (cooling, controls, fire system)? Alerting when exceeding normal range?	PREFERRED	Yes. Auxiliary power metered via dedicated sub-meter or CT clamp (included in BoM). Broken down by subsystem: HVAC/cooling, controls/UPS, fire suppression status. KPI: auxiliary consumption as % of total energy throughput. Alerting: WARNING when auxiliary > base × 120%; CRITICAL when > base × 150 %; typically indicates cooling fault or door heater malfunction.
16.8	Degradation tracking dashboard: Is there a dedicated degradation dashboard showing actual SOH vs. warranty curve, cycle count, throughput, and calendar aging breakdown? Available at container, park, and portfolio level? Can it generate reports for warranty claims?	REQUIRED	Yes. Dedicated SOH/degradation dashboard: actual SOH % vs. OEM warranty curve (configurable per OEM); cumulative cycle count; total energy throughput MWh; calendar age months; ambient temperature histogram. Available at: BMU cell string, container, park, group, and portfolio. Warranty claim report: one-click PDF export with measured data vs. warranty threshold; timestamped chain-of-custody log included.
16.9	Automated reporting: What reporting cadence is supported? (Target: monthly performance report within 10 business days, quarterly health report within 15 business days, annual review within 30 days). Are reports generated automatically or manually?	REQUIRED	Yes, fully automated. Monthly performance report: auto-generated by 5th business day of following month (T+5). Quarterly health report: auto-generated by 10th business day after quarter-end. Annual review report: auto-generated by January 15th. All reports emailed to named recipients and archived in SCADA Global portal. Manual on-demand report generation also available; custom date ranges supported.
16.10	Market bid archive: Does the EMS archive all market bid submissions and outcomes? Are bids stored with timestamp, volume, price, and acceptance status? What is the retention period?	PREFERRED	Yes. All market bid submissions stored with: UTC timestamp, service type, bid volume (MW), bid price (€/MWh), acceptance flag, delivery outcome, and settlement delta. Retention: 10 years (aligned with typical market dispute window). Exportable as CSV/Excel; filterable by date, service, and outcome. Accessible to Aggregator role via scoped API endpoint.

17. FLEET MANAGEMENT & MULTI-SITE SCADA

#	Requirement	Priority	Response
17.1	Master SCADA dashboard: Can Lighthief (as EPC/O&M provider) have a single master dashboard across all 51 parks, spanning all 3 groups and 6 standalone parks?	REQUIRED	Yes. SCADA Global "Lighthief Master" view aggregates all 51 parks across all groups and standalone sites on a single GIS-based dashboard. KPIs shown: aggregate MW, MWh, SOC%, alarms count, availability, and revenue. Drill-down from fleet → group → park → container → BMU. Real-time and historical views.
17.2	Group-level SCADA: Can each client group have its own SCADA Global instance providing consolidated view across their parks only?	REQUIRED	Yes. Each client group (ABIO, Synenergia, standalone) can have a dedicated SCADA Global tenant with visibility scoped to their parks only. Data isolation at tenant level: Group A cannot see Group B data. Multi-tenant architecture with per-group branding available. Number of group portals: unlimited.
17.3	Aggregator access: Can third-party aggregators (ABIO, Synenergia) have read-only access to their respective group's SCADA data?	REQUIRED	Yes. Aggregator role: scoped read access to their group's live telemetry (SOC, P, Q, availability) and schedule endpoint (write dispatch schedule). Access via REST API with JWT token, or optionally via direct SCADA Global dashboard login with Aggregator role. Data scope strictly limited to contracted group; no cross-group visibility.
17.4	Consolidated alarms: Can Lighthief receive consolidated alarm/fault notifications for all 51 parks in a single interface?	REQUIRED	Yes. Unified alarm manager in SCADA Global Master view. Alarms from all 51 parks filtered, deduplicated, and prioritised (CRITICAL / WARNING / INFO). Notification channels: email, SMS, Microsoft Teams webhook, PagerDuty. On-call escalation tree configurable: primary → secondary → management. Alarm suppression rules for planned maintenance windows.
17.5	Portfolio-wide reporting: Can portfolio-wide performance reports be generated across all groups?	REQUIRED	Yes. Portfolio-wide reports: monthly, quarterly, and annual; covering all 51 parks in aggregate and individual breakdown. Metrics: availability, RTE, revenue by stream, SOH/degradation, alarm counts, response compliance. Report generated automatically; also available on-demand. Export: PDF, Excel, CSV.
17.6	Scalability: How does the EMS handle capacity expansion? (a) Can additional containers/PCS be added to an existing site without full re-commissioning? (b) What is the maximum number of PCS/containers a single edge	REQUIRED	Designed for horizontal scalability. (a) Additional containers/PCS added via configuration (YAML/JSON device template) — no full re-commissioning; partial outage only for new device commissioning (≤4 h per container). (b) Single edge controller: up to 20 PCS units / 40 containers depending on comms load. Larger sites use aggregated local Modbus concentrator. (c) SCADA licenses are per-site, not per-

	controller can manage? (c) Does the SCADA license need upgrade for capacity additions?		device — no upgrade needed for capacity additions at same site.
17.7	SCADA architecture: Is the SCADA Global deployed on-premises, cloud-based, or hybrid? What server hardware is required? What is the hosting model?	REQUIRED	Hybrid — cloud default with on-premises option. Default: dedicated GPU-cluster cloud-hosted (or: AWS/Azure/GCP; customer choice) using Docker + Kubernetes. On-premises option: bare-metal or VMware ESXi. Minimum server spec (on-prem, 51 parks): 8-core CPU, 32 GB RAM, 2 TB NVMe SSD, 10 GbE NIC. Cloud: ≈8 vCPU / 32 GB RAM per cluster; storage scales automatically. Monthly cloud hosting cost included in recurring fee tier.

18. COMMERCIAL & LICENSING MODEL

18.1 Per-Site Pricing

Cost Component	Small (1–3 MW)	Medium (4–8 MW)	Large (10–15 MW)	XL (20–30 MW)
EMS Hardware + Software	€12,000 – €18,000	€18,000 – €28,000	€28,000 – €42,000	€42,000 – €65,000
Remote Configuration & Preparation	€3,000 – €4,500	€4,000 – €6,000	€5,500 – €8,000	€7,000 – €10,000
On-site Installation + Training	€5,000 – €8,000	€8,000 – €12,000	€12,000 – €18,000	€18,000 – €28,000
SCADA Local (per park)	€3,500 – €5,500	€5,000 – €8,000	€7,500 – €12,000	€11,000 – €18,000
Commissioning & DSO Testing	€4,500 – €7,000	€7,000 – €10,000	€10,000 – €16,000	€15,000 – €25,000

18.2 Fleet / Group-Level Pricing

Cost Component	Price
SCADA Global (per group, one-time)	€18,000 – €28,000 (covers all parks within the group)
Master Dashboard / Fleet Portal (one-time)	€22,000 – €35,000 (single instance covering Lighthief's full 51-park portfolio)
SCADA Global server hardware (if on-premises)	€8,000 – €15,000 per server node; dual-node HA configuration recommended (€16K–€30K). Cloud hosting eliminates this cost.

18.3 Annual / Recurring Costs

Cost Component	Annual Cost
EMS software subscription / license renewal (per park)	€3,500 – €8,000 / year (size-dependent; portfolio discount applied to all parks)
SCADA Local maintenance (per park)	€1,200 – €2,500 / year (hardware support + software updates)
SCADA Global maintenance (per group)	€4,000 – €7,000 / year per group (Lighthief Master Portal: €8,000–€12,000 / year)
Cloud / remote monitoring (per park or portfolio)	Included in EMS subscription for cloud-hosted option. On-premises: €500–€1,200 / park / year for data

	pipeline hosting. Portfolio flat rate available on request.
Remote support / helpdesk	€1,500 – €3,000 / park / year (24/7 critical response tier); standard tier (€800–€1,500 / year). Portfolio bundle pricing available.
Software updates included in subscription?	Yes — fully included. All feature releases, security patches, and regulatory updates (e.g., DSO interface changes, new market rules) included in annual subscription at no additional cost. Major version upgrades may require on-site visit (billed separately if hardware changes needed).

18.4 Volume & Contract Questions

Question	Response
Is there a volume discount for a 51-park portfolio order?	Yes. 51-park portfolio: 15–20% discount on all per-park capex items. Discount applied to each park regardless of size. Single MSA for all 51 parks with unified commercial terms.
Is there a discount for multi-year subscription commitment (3, 5, or 10 years)?	Yes. 3-year commitment: 5% discount. 5-year: 10%. 10-year: 15%. Discounts apply to annual subscription fees and maintenance items. Price escalation cap: CPI or 3% max per year, whichever is lower.
Can standalone parks receive discounted pricing if ordered as part of the portfolio?	Yes. All 51 parks (regardless of group affiliation) benefit from portfolio pricing when ordered under the same MSA. Standalone parks treated as full portfolio members for discount calculation.
Is the EMS software licensed perpetually or subscription-based? If subscription expires, what functionality remains?	Subscription-based. On expiry: read-only SCADA (monitoring continues), historical data accessible, local edge controller dispatch continues executing last loaded schedule. Remote control and optimisation features suspended. 30-day grace period after expiry before read-only mode. Source code escrow ensures continuity.
Is there an escrow arrangement for EMS source code?	Yes, available under Enterprise MSA. Source code deposited with Escrow Europe GmbH (or equivalent) upon contract signing. Release conditions: supplier insolvency, cessation of business, or material breach with failure to cure within 60 days. Escrow annual fee borne by supplier.
What payment terms are offered (advance, milestone-based, etc.)?	Milestone-based. 30% on contract signing, 40% on hardware delivery to site, 30% on commissioning acceptance (SAT signed-off). Annual subscription invoiced at the start of each contract year. Net 30

payment terms. Progress billing available for large multi-phase rollouts.

19. SUPPORT, WARRANTY & SLA

#	Question	Response
19.1	Hardware warranty period?	2 years from date of FAT sign-off. Covers PPC/RTU hardware, networking equipment, cabinet, and supplied sensors. Warranty replacement parts shipped within 5 business days.
19.2	Software warranty / bug fix period?	2 years from commissioning date. Critical bugs: patched within 72 h. High severity: 7 days. Bug fix releases provided at no cost during warranty period. Warranty extended automatically while active subscription in place.
19.3	Is the warranty extendable (1, 3, 5, 10 years)? Under what conditions?	Yes. Extended warranty available: 1, 3, or 5 years. Hardware extended warranty requires annual preventive maintenance inspection (remote or on-site). Software warranty is automatically covered while subscription is active. 10-year option available for parks with active subscription + annual inspection contract.
19.4	Remote support availability hours? (24/7 or business hours?)	24/7 for critical faults (production-impacting events). Standard support: business hours Mon–Fri 08:00–18:00 CET. After-hours critical support included in standard MSA; 24/7 full helpdesk access available as premium tier add-on.
19.5	Response time SLA for critical faults? For non-critical issues?	Critical (P1 — production loss): 2-hour initial response; 8-hour resolution target. High (P2 — degraded operation): 4-hour initial response; 24-hour resolution. Medium (P3): Next business day acknowledgement; 5-business-day resolution. Low (P4): 10 business days. SLA credits apply for breach.
19.6	Is on-site support available in Cyprus? If not, what is the mobilization time?	Via certified partners in Cyprus / Eastern Mediterranean. Mobilisation from Poland HQ: 48–72 hours. Local partner network established during project commissioning phase. On-site emergency response: flights + logistics arrangement within 24 hours of P1 escalation. Remote diagnosis/resolution covers >90% of faults.
19.7	Is operator training included? How many days? On-site or remote?	Yes, included. 3 days standard: Day 1 — SCADA operation (remote), Day 2 — EMS/optimisation module (remote), Day 3 — on-site at first commissioned park (hardware + integration). Lighthief O&M team: additional training package for all 51 parks available (2-day refresher per group). LMS video training library included, accessible 24/7.
19.8	What documentation is provided (user manual, integration guide, API)	Full documentation package: (1) User & Operator Manual (EN/PL, PDF + online). (2) Integration Guide: Kehua PCS Modbus map, Linyang BMS comms, IEC-104

	documentation, network architecture guide)?	point list template. (3) REST API Reference (OpenAPI 3.0 / Swagger). (4) Network Architecture Guide with firewall rule templates. (5) Commissioning Checklist & SAT test protocol. (6) Cybersecurity Configuration Guide. All documents delivered at commissioning; version-controlled in customer portal.
19.9	What is the typical implementation timeline from contract signing to first site go-live?	6–8 weeks per site from contract signing: Week 1–2: hardware order + remote pre-configuration. Week 3–4: hardware delivery + site installation by local electrician. Week 5–6: remote commissioning, IEC-104 DSO testing, FAT. Week 7–8: SAT, operator training, go-live. Parallel commissioning of multiple sites: 4–6 sites simultaneously achievable; 51-site rollout: 6–12 months depending on site readiness.
19.10	Is FAT (Factory Acceptance Test) and SAT (Site Acceptance Test) support included?	Yes, both included at no additional charge for standard deployments. FAT: pre-configured hardware tested remotely against agreed test protocol before shipment (Kehua PCS Modbus, Linyang BMS, IEC-104 simulation, fail-safe logic, LFSM-O, P(V)). SAT: on-site or remote (engineer connected live) verification of live DSO IEC-104 connection, full dispatch cycle, alarm escalation, failsafe test. FAT/SAT test reports provided as signed PDF documents.

20. COMPANY QUALIFICATIONS & REFERENCES

Information	Response
Company legal name	Diffused Energy North sp. z o.o.
Country / HQ location	Poland — Bydgoszcz. EU-registered company.
Years in BESS EMS business	7 years in BESS EMS development; energy management software since 2015; BESS-specific EMS platform (diffusedEMS) actively developed since 2020.
Total BESS projects delivered (MW / MWh)	To be confirmed at RFP stage with client reference list. Pilot and commercial projects: >50 MW EMS under management or delivered; full detail provided in company profile document appended to this response.
Total fleet under active management (MW / MWh)	Details to be confirmed with NDA at RFP stage. Current platform supports multi-site portfolio management; active fleet data provided upon formal RFP engagement.
EU / Cyprus / Mediterranean references	Reference projects in Poland (Baltic coast BESS + wind hybrid), Orange (solar + BESS, grid balancing). Cyprus/Mediterranean: this portfolio represents our first direct Cyprus project; regional subcontractor experience in Eastern Mediterranean available. Reference contacts provided with RFP response.
Kehua PCS integration experience? (BCS-C-HUD series)	Yes. Kehua BCS-C-HUD series integrated via Modbus RTU (RS-485) and Modbus TCP. Full register map implemented: control registers (active/reactive power setpoints, on/off), status registers (DC voltage, current, temperature, fault codes), alarm words. Integration tested on BCS-C-HUD-1000 and BCS-C-HUD-500 units.
Linyang BMS integration experience?	Yes. Linyang Energy 3-level BMS (BMU / BCMU / BAMS) integrated via CAN bus (cell-level) and Modbus TCP (BCMU/BAMS). Data mapped: cell voltages, temperatures, SOC, SOH, alarm words per string and pack level. Power Atlantic 5.015 MWh container architecture (96 BMUs, 3 BCMU, 1 BAMS) fully supported.
IEC 60870-5-104 integration experience? (How many DSOs?)	Yes — 3+ DSO integrations completed. IEC 60870-5-104 implemented with full ASDU type support (C_DC, C_SC, M_ME, M_SP, M_IT). Integrated with: PSE S.A. (Poland), Enea Operator (Poland), pilot with ENTSO-E test environment. TSOC (Cyprus) integration planned for this project using standard ASDUs specified in tender requirements.
Largest single BESS site managed (MW / MWh)?	15 MW / 30 MWh (solar + BESS hybrid, Poland). EMS platform validated at XL site tier. Reference details provided under NDA at RFP stage.

Largest fleet / portfolio managed (total MW / MWh)?	Multi-site portfolio: >50 MW / >100 MWh under single SCADA Global instance. 51-park Cyprus portfolio (249 MW / 882 MWh) would represent our largest deployment — platform designed and scalability-tested for this scale.
Number of full-time engineering staff dedicated to EMS	7 FTE engineers dedicated to EMS: 2 backend / platform engineers, 1 controls / integration engineers, 1 data science / ML engineers, 1 frontend / UX engineers, 1 DevOps / cybersecurity engineer. Support team: (1x Admin - scalable). Senior management: 1 (CEO/CTO/Architect).
Company certifications (ISO 9001, ISO 27001, IEC 62443, etc.)	Current: ISO 9001:2015 (quality management); IEC 62443 SL1 self-assessment completed. In progress: ISO 27001 (target Q3 2026); IEC 62443 SL2 certification (target H1 2027). Software development: GitLab CI/CD + automated testing pipeline; SOC 2 Type I assessment planned 2026.
Can you provide 3 reference contacts from similar BESS EMS projects?	Yes. Three reference contacts from BESS EMS projects provided in the confidential appendix to this RFI response. References include EPC/O&M managers from Poland as well as USA and Canada projects. Lighthief may contact references directly; introductions arranged upon request. Additional references available under NDA at RFP stage.

21. RESPONSE INSTRUCTIONS

Please submit your response to:

Email: office@lighthief.com

Subject: RFI Response — EMS Cyprus BESS Portfolio (LCY-RFI-EMS-2026-004)

Deadline: [To be confirmed]

Response deliverables:

1. Completed RFI questionnaire (this document, all response fields filled)
2. Company profile / capabilities brochure
3. Reference project list with contact details
4. Technical architecture proposal (system diagram per site size category)
5. Indicative pricing (budget level acceptable at RFI stage)
6. Detailed Bill of Materials per site size category (Small / Medium / Large / XL)
7. Sample IEC 60870-5-104 point list for BESS site
8. Cybersecurity compliance statement (NIS2 / IEC 62443)
9. Implementation timeline proposal

Important Notes:

- All proposals must be valid for a minimum of 90 days from submission date
- Pricing must be in EUR (€)
- This RFI does not constitute a commitment to purchase — it is an information-gathering exercise to evaluate EMS providers for our portfolio
- Lighthief reserves the right to issue a formal RFP to shortlisted respondents
- All information provided will be treated as confidential

21.1 Evaluation Criteria

Criteria	Weight
Technical capability (EMS features, optimization, grid compliance)	30%
Integration experience (Kehua PCS, Linyang BMS, IEC-104)	20%
Fleet management & scalability (51 parks, multi-group SCADA)	15%
Cybersecurity & NIS2 compliance	10%
Commercial terms (capex, annual costs, licensing model)	15%
Support, warranty & SLA (availability, response time, Cyprus presence)	10%

TOTAL	100%
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QUESTIONNAIRE SUMMARY

Section	Questions	Focus Area
1. Introduction	—	Portfolio context & BESS specifications
2. System Architecture	14	Hardware scope & BoM
3. PPC & Grid Code	10	Power plant control & EN 50549-2
4. Communication Protocols	10	IEC-104, Modbus, DSO integration
5. Battery Monitoring	6	Cell-level data, SOC/SOH estimation
6. Degradation Analytics	5	Cycle counting, calendar aging, EOL forecast
7. Thermal & Safety	5	Thermal runaway, gas detection, fire system
8. Fault Detection	4	Classification, predictive maintenance, RCA
9. Data Infrastructure	6	Logging resolution, retention, APIs
10. EMS Optimization	8	Multi-objective, degradation-aware dispatch
11. Market Integration	5	Day-ahead, intraday, ancillary services
12. Forecast Integration	5	Price, RES, load forecasting
13. Remote Dispatch	6	Fail-safe, priority hierarchy, autonomy
14. Redundancy	5	Server, communication, UPS, failover
15. Cybersecurity	8	RBAC, encryption, NIS2, IEC 62443
16. KPI Monitoring	10	Availability, RTE, revenue, degradation
17. Fleet Management	7	Multi-site SCADA, aggregator access
18. Commercial	16	Capex, opex, licensing, discounts
19. Support & Warranty	10	SLA, training, documentation, timeline
20. Company Qualifications	14	Experience, references, certifications
TOTAL QUESTIONS	154	

Prepared by: Lighthief Cyprus Ltd

28 October Avenue 249, Lophitis Business Center 1, Office
201

3035 Limassol, Cyprus

Email: office@lighthief.com | Tel: +357 99 164 158

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